

ASSIGNMENT FEEDBACK FORM

Student	Mohammed Abdelhakeem Ahmed Al-maaitah - 22210029 - 1		
HTU Course Title and No.	10204351 - Natural Language Processing		
BTEC Unit, Title, and ID	-		
Assignment Number	1	Assessor	Yara Alharahsheh
Submission Date	Jan 25, 2026	Date Received 1st submission	Jan 25, 2026
Re-submission Date		Date Received 2nd submission	

Ongoing formative feedback from assessor:

- **Attendance:** Very Good
- **Lab / In-class Activity:** Good
- **One-to-one Formative Discussion:** Very Good

Assessor feedback for summative assessment:

General Feedback

During the in-class examination, Mohammad showed an ability of understanding of natural language processing concepts, including issues related to bias, as well as knowledge of linguistic subfields. He has an understanding of word representation techniques and showed awareness of the theoretical aspects of both traditional NLP models. However, it is recommended that he place greater emphasis on the critical assessment of the technical implementation of word representation techniques and neural network-based NLP models.

In the project on multiclass Arabic news text classification, Mohammad identified and implemented three word representation techniques: Traditional (TF-IDF), Independent (FastText), and Dependent (BERT), noting their general characteristics and observing performance trends, with TF-IDF showing consistently poor results, FastText providing notable improvements, and BERT achieving the highest accuracy. However, the comparison lacked explicit, structured contrast of their underlying mechanisms, dimensionality, or specific contextual limitations supported by detailed accuracy metrics. He tested several NLP models, including Decision Tree, LSTM, GRU, and a fine-tuned BERT classifier, noting that LSTM was sensitive to the choice of embedding, GRU offered lower computational complexity, and BERT delivered the best overall performance. Yet the analysis remained descriptive rather than analytical; architectural strengths and weaknesses were not systematically discerned, and performance rankings were not clearly quantified or tied to model design. Mohammad also explored three Arabic stemming methods, finding that Snowball stemming generally yielded the best results, especially with FastText and BERT, while ISRI stemming sometimes harmed performance. However, this evaluation was confined to classification accuracy without linking the preprocessing choices to real-world deployment scenarios—such as processing speed, maintainability, or suitability for different application environments; and therefore did not critically assess their practical effectiveness. Lastly, while he acknowledged that TF-IDF is suitable for lightweight systems and BERT demands more resources, the real-world applicability of the word representation techniques was not evaluated against specific use-case constraints (e.g., latency, budget, scalability) or operational trade-offs. The project reported experimental outcomes but did not translate them into actionable, scenario-based recommendations for solving actual Arabic NLP problems in production.

Strength of Performance

The student was able to:

- Explore the historical evolution, diverse techniques, and wide-ranging applications of natural language processing, while analyzing associated bias concerns.
- Explore a range of linguistic subfields to analyze and describe diverse aspects of language.
- Understand word representation techniques employed in text encoding for natural language processing tasks.
- Apply diverse word representation techniques to address real-life applications.
- Understand the traditional NLP models and their applications across various domains.
- Implement traditional NLP models to address real-life applications.
- Acquire comprehension of neural language models and their applications across diverse natural language processing tasks.
- Implement neural language models to solve real-life applications effectively.
- Implement preprocessing methods and relevant packages to address real-life applications within natural language processing.
- Critically assess the technical implementation of various word representation techniques for natural language processing tasks.
- Compare and contrast different word representation techniques used in text encoding for natural language processing tasks.
- Conduct comparative analysis of different NLP models to discern their respective strengths and weaknesses.

Limitation of Performance

The student was unable to:

- Evaluate the effectiveness of the selected preprocessing methods in solving real-world applications.
- Evaluate the effectiveness of word representation techniques in addressing real-life applications.

Grade: M	Assessor Signature: <i>Yara Emad Abdelghafour Alharahsheh</i>	Date: Jan 26, 2026
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Resubmission Feedback (if required):

Grade:	Assessor Signature:	Date:
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Criteria (To be filled before resubmission)

P1 ☒	P2 ☒	P3 ☒	P4 ☒	P5 ☒	P6 ☒	P7 ☒	<u>Final Grade</u>
P8 ☒	M1 ☒	M2 ☒	M3 ☒	M4 ☒	D1 ☐	D2 ☐	<u>M</u>

Assessor Declaration:

I declare that the work submitted for assessment has been carried out without assistance other than that which is acceptable according to the rules of the specification. I certify that to the best of my knowledge the evidence submitted for this assignment is the student's own. The student has clearly referenced any sources, and any AI tools used in the work. I have not solely used AI to mark student's work. I understand that false declaration is a form of malpractice.

Assessor Name: Yara Alharahsheh Assessor Signature: Yara Alharahsheh

Student Declaration:

I certify that the formative and summative assessments for this assignment have been fully explained and understood by me, I also do understand that the grade above is simply a recommendation that could later be changed during any of the verification processes.

Student Name:

Student Signature:

Date: